

Benchmarking Results (terminal screenshots)

Machine A - Laptop

- **CPU1:** 13th Gen Intel(R) Core(TM) i5-13420H
 - **Cores:** 8 cores, 12 logical processors
- **GPU1:** NVIDIA GeForce RTX 4050
 - **VRAM:** 6GB
 - **CUDA version:** 12.8

Machine B - Workstation

- **CPU2:** Intel(R) Xeon(R) w5-3435X
 - **Cores:** 16 cores, 32 logical processors
- **GPU2:** NVIDIA RTX 4000 Ada Generation
 - **VRAM:** 20GB
 - **CUDA version:** 12.8

FP32 (CPU1)

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/640/yolov8n_640.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 23.9696
Model size (on disk): 12.26 MiB (12,851,087 bytes)
mAP@IoU=0.50 : 0.681449
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

```
mAP@IoU=0.50 : 0.681449
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/640/yolov8s_640.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 10.3011
Model size (on disk): 42.79 MiB (44,869,881 bytes)
mAP@IoU=0.50 : 0.709116
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/640/yolov8m_640.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 1.9216
Model size (on disk): 99.00 MiB (103,809,586 bytes)
mAP@IoU=0.50 : 0.723569
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/320/yolov8n_320.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 64.1732
Model size (on disk): 12.14 MiB (12,725,005 bytes)
mAP@IoU=0.50 : 0.521438
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

dels\640\yolov8s_320.onnx
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/320/yolov8s_320.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 35.7109
Model size (on disk): 42.67 MiB (44,743,799 bytes)
mAP@IoU=0.50 : 0.614547
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/320/yolov8m_320.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 2.8104
Model size (on disk): 98.88 MiB (103,683,476 bytes)
mAP@IoU=0.50 : 0.660944
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/1280/yolov8n_1280.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 5.9072
Model size (on disk): 12.74 MiB (13,355,362 bytes)
mAP@IoU=0.50 : 0.725778

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/1280/yolov8s_1280.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 2.5601
Model size (on disk): 43.27 MiB (45,374,156 bytes)
mAP@IoU=0.50 : 0.761152

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P32_onnx_models/1280/yolov8m_1280.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 0.9990
Model size (on disk): 99.48 MiB (104,313,941 bytes)
mAP@IoU=0.50 : 0.757325
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

FP16 quantized (CPU1)

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/640/yolov8n_640_fp16.onnx --device cpu --input-size 640 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
  inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
  return inter_area / union_area
FPS: 13.5533
Model size (on disk): 6.14 MiB (6,436,157 bytes)
mAP@IoU=0.50 : 0.681476
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/640/yolov8s_640_fp16.onnx --device cpu --input-size 640 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 6.7362
Model size (on disk): 21.41 MiB (22,445,539 bytes)
mAP@IoU=0.50 : 0.709287

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/640/yolov8m_640_fp16.onnx --device cpu --input-size 640 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 3.2922
Model size (on disk): 49.51 MiB (51,917,808 bytes)
mAP@IoU=0.50 : 0.723501
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

```

320 input size

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/320/yolov8n_320_fp16.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 61.7969
Model size (on disk): 6.08 MiB (6,373,145 bytes)
mAP@IoU=0.50 : 0.520598
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/320/yolov8s_320_fp16.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 21.8552
Model size (on disk): 21.35 MiB (22,382,527 bytes)
mAP@IoU=0.50 : 0.614515
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/320/yolov8m_320_fp16.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 11.0203
Model size (on disk): 49.45 MiB (51,854,796 bytes)
mAP@IoU=0.50 : 0.663453
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

```

1280 input size

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/1280/yolov8n_1280_fp16.onnx --device cpu --input-size 1280 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 5.1509
Model size (on disk): 6.38 MiB (6,688,163 bytes)
mAP@IoU=0.50 : 0.724195

```

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/1280/yolov8s_1280_fp16.onnx --device cpu --input-size 1280 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 2.2968
Model size (on disk): 21.65 MiB (22,697,545 bytes)
mAP@IoU=0.50 : 0.759581

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/F
P16_quantized_models/1280/yolov8m_1280_fp16.onnx --device cpu --input-size 1280 --data-root imagedata
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:34: RuntimeWarning: overflow encountered in scalar mult
iply
    inter_area = (x2 - x1).clip(min=0) * (y2 - y1).clip(min=0)
C:\Users\rjrao\Desktop\P_Det\code_files\postprocessing.py:66: RuntimeWarning: invalid value encountered in scalar
divide
    return inter_area / union_area
FPS: 0.9149
Model size (on disk): 49.75 MiB (52,169,814 bytes)
mAP@IoU=0.50 : 0.755405

```

INT8 Static Quantized (CPU1)

320 input size

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/I
NT8_static_quantized_models/320/yolov8n_320_static_quantized.onnx --device cpu --input-size 320 --data-root imag
data
FPS: 16.1451
Model size (on disk): 3.31 MiB (3,474,227 bytes)
mAP@IoU=0.50 : 0.497935

NT8_static_quantized_models/320/yolov8s_320_static_quantized.onnx --device cpu --input-size 320 --data-root image
data
FPS: 8.5828
Model size (on disk): 11.02 MiB (11,554,684 bytes)
mAP@IoU=0.50 : 0.564415

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/I
NT8_static_quantized_models/320/yolov8m_320_static_quantized.onnx --device cpu --input-size 320 --data-root image
data
FPS: 8.2609
Model size (on disk): 25.22 MiB (26,448,568 bytes)
mAP@IoU=0.50 : 0.633234
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

```

640 input size

```

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/I
NT8_static_quantized_models/640/yolov8n_640_static_quantized.onnx --device cpu --input-size 640 --data-root image
data
FPS: 15.5196
Model size (on disk): 3.43 MiB (3,600,269 bytes)
mAP@IoU=0.50 : 0.653009

```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/INT8_static_quantized_models/640/yolov8s_640_static_quantized.onnx --device cpu --input-size 640 --data-root image data
FPS: 5.7725
Model size (on disk): 11.14 MiB (11,680,726 bytes)
mAP@IoU=0.50 : 0.691776
```

```
mAP@IoU=0.50 : 0.691776
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/INT8_static_quantized_models/640/yolov8m_640_static_quantized.onnx --device cpu --input-size 640 --data-root image data
FPS: 2.6777
Model size (on disk): 25.34 MiB (26,574,624 bytes)
mAP@IoU=0.50 : 0.713427
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/INT8_static_quantized_models/1280/yolov8n_1280_static_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 4.0739
Model size (on disk): 3.91 MiB (4,104,411 bytes)
mAP@IoU=0.50 : 0.704670
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/INT8_static_quantized_models/1280/yolov8s_1280_static_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 1.5106
Model size (on disk): 11.62 MiB (12,184,868 bytes)
mAP@IoU=0.50 : 0.733785
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models/yolo_onnx_models/INT8_static_quantized_models/1280/yolov8m_1280_static_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 0.7107
Model size (on disk): 25.82 MiB (27,078,806 bytes)
mAP@IoU=0.50 : 0.759759
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

INT8 Dynamic quantized (CPU1)

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8n_320_dynamic_quantized.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 66.2173
device=cpu
Model size (on disk): 12.14 MiB (12,725,187 bytes)
mAP@IoU=0.50 : 0.521438
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8s_320_dynamic_quantized.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 12.2797
device=cpu
Model size (on disk): 42.67 MiB (44,744,007 bytes)
mAP@IoU=0.50 : 0.614228
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8m_320_dynamic_quantized.onnx --device cpu --input-size 320 --data-root imagedata
FPS: 13.7941
device=cpu
Model size (on disk): 98.88 MiB (103,683,732 bytes)
mAP@IoU=0.50 : 0.660906
```

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8n_640_dynamic_quantized.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 15.0983
device=cpu
Model size (on disk): 12.26 MiB (12,851,269 bytes)
mAP@IoU=0.50 : 0.679877

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8s_640_dynamic_quantized.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 9.8318
device=cpu
Model size (on disk): 42.79 MiB (44,870,089 bytes)
mAP@IoU=0.50 : 0.707885

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8m_640_dynamic_quantized.onnx --device cpu --input-size 640 --data-root imagedata
FPS: 3.6132
device=cpu
Model size (on disk): 99.00 MiB (103,809,842 bytes)
mAP@IoU=0.50 : 0.723569
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8n_1280_dynamic_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 5.6409
device=cpu
Model size (on disk): 12.74 MiB (13,355,544 bytes)
mAP@IoU=0.50 : 0.725776

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8s_1280_dynamic_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 2.4696
device=cpu
Model size (on disk): 43.27 MiB (45,374,364 bytes)
mAP@IoU=0.50 : 0.760904

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8m_1280_dynamic_quantized.onnx --device cpu --input-size 1280 --data-root imagedata
FPS: 0.7726
device=cpu
Model size (on disk): 99.48 MiB (104,314,197 bytes)
mAP@IoU=0.50 : 0.757325
```

FP32 (GPU1)

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_models\320\yolov8n_320.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 68.4923
device=gpu
Model size (on disk): 12.14 MiB (12,725,005 bytes)
mAP@IoU=0.50 : 0.521454
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_models\320\yolov8s_320.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 57.3142
device=gpu
Model size (on disk): 42.67 MiB (44,743,799 bytes)
mAP@IoU=0.50 : 0.614213

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_models\320\yolov8m_320.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 28.9642
device=gpu
Model size (on disk): 98.88 MiB (103,683,476 bytes)
mAP@IoU=0.50 : 0.660919
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_models\640\yolov8n_640.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 39.4660
device=gpu
Model size (on disk): 12.26 MiB (12,851,087 bytes)
mAP@IoU=0.50 : 0.679865

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_models\640\yolov8s_640.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 25.7842
device=gpu
Model size (on disk): 42.79 MiB (44,869,881 bytes)
mAP@IoU=0.50 : 0.707891

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_models\640\yolov8m_640.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 13.9254
device=gpu
Model size (on disk): 99.00 MiB (103,809,586 bytes)
mAP@IoU=0.50 : 0.723554
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

1280 input size

```
mAP@IoU=0.50 : 0.725534
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_models\1280\yolov8n_1280.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 14.8924
device=gpu
Model size (on disk): 12.74 MiB (13,355,362 bytes)
mAP@IoU=0.50 : 0.725816

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_models\1280\yolov8s_1280.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 9.2814
device=gpu
Model size (on disk): 43.27 MiB (45,374,156 bytes)
mAP@IoU=0.50 : 0.760830
python code_files/main_execution.py --model models\yolo_onnx_models\FP32_onnx_models\1280\yolov8m_1280.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 5.1672
device=gpu
Model size (on disk): 99.48 MiB (104,313,941 bytes)
mAP@IoU=0.50 : 0.757285
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

INT8 Static quantization (GPU1)

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8n_320_static_quantized.onnx --device gpu --input-size 320 --data-root imagedata
2026-05-03 22:16:42.9800676 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy nodes are added to the graph main_graph for CUDAExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 52.0796
device=gpu
Model size (on disk): 3.31 MiB (3,474,227 bytes)
mAP@IoU=0.50 : 0.491260
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8s_320_static_quantized.onnx --device gpu --input-size 320 --data-root imagedata
2026-05-03 22:17:08.8398341 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy nodes are added to the graph main_graph for CUDAExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 42.9500
device=gpu
Model size (on disk): 11.02 MiB (11,554,684 bytes)
mAP@IoU=0.50 : 0.564701
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8m_320_static_quantized.onnx --device gpu --input-size 320 --data-root imagedata
2026-05-03 22:17:23.7182927 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Memcpy nodes are added to the graph main_graph for CUDAExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 24.4121
device=gpu
Model size (on disk): 25.22 MiB (26,448,568 bytes)
mAP@IoU=0.50 : 0.626899
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

640 input size

```
mAP@IoU=0.50 : 0.626899
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8n_640_static_quantized.onnx --device gpu --input-size 640 --data-root imagedata
2026-05-03 22:19:16.5935512 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy nodes are added to the graph main_graph for CUDAExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 30.5967
device=gpu
Model size (on disk): 3.43 MiB (3,600,269 bytes)
mAP@IoU=0.50 : 0.658708
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8s_640_static_quantized.onnx --device gpu --input-size 640 --data-root imagedata
2026-05-03 22:19:29.6263499 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy nodes are added to the graph main_graph for CUDAExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 22.3620
device=gpu
Model size (on disk): 11.14 MiB (11,680,726 bytes)
mAP@IoU=0.50 : 0.684866
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8m_640_static_quantized.onnx --device gpu --input-size 640 --data-root imagedata
2026-05-03 22:19:40.5292893 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Memcpy nodes are added to the graph main_graph for CUDAExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 11.8089
device=gpu
Model size (on disk): 25.34 MiB (26,574,624 bytes)
mAP@IoU=0.50 : 0.708243
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8n_1280_static_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
2026-05-03 22:20:23.6149767 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 11.5545
device=gpu
Model size (on disk): 3.91 MiB (4,104,411 bytes)
mAP@IoU=0.50 : 0.710115
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8s_1280_static_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
2026-05-03 22:20:49.3277053 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Memcpy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 8.1129
device=gpu
Model size (on disk): 11.62 MiB (12,184,868 bytes)
mAP@IoU=0.50 : 0.731770

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8m_1280_static_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
2026-05-03 22:21:09.9473603 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Memcpy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 4.0203
device=gpu
Model size (on disk): 25.82 MiB (27,078,806 bytes)
mAP@IoU=0.50 : 0.747494
```

INT8 dynamic quantization (GPU1)

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8n_320_dynamic_quantized.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 68.7856
device=gpu
Model size (on disk): 12.14 MiB (12,725,187 bytes)
mAP@IoU=0.50 : 0.521454

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8s_320_dynamic_quantized.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 48.6796
device=gpu
Model size (on disk): 42.67 MiB (44,744,007 bytes)
mAP@IoU=0.50 : 0.614213

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8m_320_dynamic_quantized.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 29.7171
device=gpu
Model size (on disk): 98.88 MiB (103,683,732 bytes)
mAP@IoU=0.50 : 0.600919
```

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8n_640_dynamic_quantized.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 38.7926
device=gpu
Model size (on disk): 12.26 MiB (12,851,269 bytes)
mAP@IoU=0.50 : 0.679865
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8s_640_dynamic_quantized.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 25.9110
device=gpu
Model size (on disk): 42.79 MiB (44,870,089 bytes)
mAP@IoU=0.50 : 0.707891
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8m_640_dynamic_quantized.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 16.7324
device=gpu
Model size (on disk): 99.00 MiB (103,809,842 bytes)
mAP@IoU=0.50 : 0.723554
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8n_1280_dynamic_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 11.3084
device=gpu
Model size (on disk): 12.74 MiB (13,355,544 bytes)
mAP@IoU=0.50 : 0.725816
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8s_1280_dynamic_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 9.8726
device=gpu
Model size (on disk): 43.27 MiB (45,374,364 bytes)
mAP@IoU=0.50 : 0.760830
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8m_1280_dynamic_quantized.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 5.1728
device=gpu
Model size (on disk): 99.48 MiB (104,314,197 bytes)
mAP@IoU=0.50 : 0.757285
```

FP16 quantization (GPU1)

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\320\yolov8n_320_fp16.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 51.4656
device=gpu
Model size (on disk): 6.08 MiB (6,373,145 bytes)
mAP@IoU=0.50 : 0.520139
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\320\yolov8s_320_fp16.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 50.7065
device=gpu
Model size (on disk): 21.35 MiB (22,382,527 bytes)
mAP@IoU=0.50 : 0.614227
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\320\yolov8m_320_fp16.onnx --device gpu --input-size 320 --data-root imagedata
FPS: 28.2521
device=gpu
Model size (on disk): 49.45 MiB (51,854,796 bytes)
mAP@IoU=0.50 : 0.659276
```

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\640\yolov8n_640_fp16.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 19.7382
device=gpu
Model size (on disk): 6.14 MiB (6,436,157 bytes)
mAP@IoU=0.50 : 0.679815

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\640\yolov8s_640_fp16.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 28.1420
device=gpu
Model size (on disk): 21.41 MiB (22,445,539 bytes)
mAP@IoU=0.50 : 0.709141

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\640\yolov8m_640_fp16.onnx --device gpu --input-size 640 --data-root imagedata
FPS: 19.2476
device=gpu
Model size (on disk): 49.51 MiB (51,917,808 bytes)
mAP@IoU=0.50 : 0.723527
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\1280\yolov8n_1280_fp16.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 10.5133
device=gpu
Model size (on disk): 6.38 MiB (6,688,163 bytes)
mAP@IoU=0.50 : 0.725673

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\1280\yolov8s_1280_fp16.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 7.8868
device=gpu
Model size (on disk): 21.65 MiB (22,697,545 bytes)
mAP@IoU=0.50 : 0.761154

(yolo_env) PS C:\Users\rjrao\Desktop\P_Det> python code_files/main_execution.py --model models\yolo_onnx_models\FP16_quantized_models\1280\yolov8m_1280_fp16.onnx --device gpu --input-size 1280 --data-root imagedata
FPS: 5.1841
device=gpu
Model size (on disk): 49.75 MiB (52,169,814 bytes)
mAP@IoU=0.50 : 0.757309
(yolo_env) PS C:\Users\rjrao\Desktop\P_Det>
```

FP32 (GPU2)

320 input size

```
-cudf-cu12-11.4.1.4 nvdata-cuda12-cu12-10.3.10.19 nvdata-nvjitlink-cu12-12.9.86
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8n_320.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 69.8035
device=gpu
Model size (on disk): 12.14 MiB (12,725,005 bytes)
mAP@IoU=0.50 : 0.518815
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8s_320.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 67.1662
device=gpu
Model size (on disk): 42.67 MiB (44,743,799 bytes)
mAP@IoU=0.50 : 0.614202

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8m_320.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 56.2887
device=gpu
Model size (on disk): 98.88 MiB (103,683,476 bytes)
mAP@IoU=0.50 : 0.660910
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8n_640.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 43.7517
device=gpu
Model size (on disk): 12.26 MiB (12,851,087 bytes)
mAP@IoU=0.50 : 0.679870
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8s_640.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 40.6903
device=gpu
Model size (on disk): 42.79 MiB (44,869,881 bytes)
mAP@IoU=0.50 : 0.707891
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8m_640.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 35.0169
device=gpu
Model size (on disk): 99.00 MiB (103,809,586 bytes)
mAP@IoU=0.50 : 0.723554
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8n_1280.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 17.9713
device=gpu
Model size (on disk): 12.74 MiB (13,355,362 bytes)
mAP@IoU=0.50 : 0.725816
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8s_1280.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 15.8469
device=gpu
Model size (on disk): 43.27 MiB (45,374,156 bytes)
mAP@IoU=0.50 : 0.760824
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8m_1280.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 13.1702
device=gpu
Model size (on disk): 99.48 MiB (104,313,941 bytes)
mAP@IoU=0.50 : 0.757285
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

FP16 quantization (GPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8n_320_fp16.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 63.8759
device=gpu
Model size (on disk): 6.08 MiB (6,373,145 bytes)
mAP@IoU=0.50 : 0.520139
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8s_320_fp16.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 57.6318
device=gpu
Model size (on disk): 21.35 MiB (22,382,527 bytes)
mAP@IoU=0.50 : 0.614337
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8m_320_fp16.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 50.3925
device=gpu
Model size (on disk): 49.45 MiB (51,854,796 bytes)
mAP@IoU=0.50 : 0.660755
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8n_640_fp16.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 34.2168
device=gpu
Model size (on disk): 6.14 MiB (6,436,157 bytes)
mAP@IoU=0.50 : 0.679815

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8s_640_fp16.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 30.1633
device=gpu
Model size (on disk): 21.41 MiB (22,445,539 bytes)
mAP@IoU=0.50 : 0.709162

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8m_640_fp16.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 30.4514
device=gpu
Model size (on disk): 49.51 MiB (51,917,808 bytes)
mAP@IoU=0.50 : 0.723568
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8n_1280_fp16.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 13.2606
device=gpu
Model size (on disk): 6.38 MiB (6,688,163 bytes)
mAP@IoU=0.50 : 0.725643

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8s_1280_fp16.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 12.4568
device=gpu
Model size (on disk): 21.65 MiB (22,697,545 bytes)
mAP@IoU=0.50 : 0.761018

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8m_1280_fp16.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 10.7555
device=gpu
Model size (on disk): 49.75 MiB (52,169,814 bytes)
mAP@IoU=0.50 : 0.757326
```

INT8 Dynamic Quantization (GPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8n_320_dynamic_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 74.1708
device=gpu
Model size (on disk): 12.14 MiB (12,725,187 bytes)
mAP@IoU=0.50 : 0.518815

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8s_320_dynamic_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 67.5562
device=gpu
Model size (on disk): 42.67 MiB (44,744,007 bytes)
mAP@IoU=0.50 : 0.614202

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8m_320_dynamic_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
FPS: 60.9990
device=gpu
Model size (on disk): 98.88 MiB (103,683,732 bytes)
mAP@IoU=0.50 : 0.660910
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8n_640_dynamic_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 45.9126
device=gpu
Model size (on disk): 12.26 MiB (12,851,269 bytes)
mAP@IoU=0.50 : 0.679870

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8s_640_dynamic_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 41.3695
device=gpu
Model size (on disk): 42.79 MiB (44,870,889 bytes)
mAP@IoU=0.50 : 0.707891

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8m_640_dynamic_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
FPS: 36.8402
device=gpu
Model size (on disk): 99.00 MiB (103,809,842 bytes)
mAP@IoU=0.50 : 0.723554
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8n_1280_dynamic_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 18.1003
device=gpu
Model size (on disk): 12.74 MiB (13,355,544 bytes)
mAP@IoU=0.50 : 0.725816
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8s_1280_dynamic_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 16.5224
device=gpu
Model size (on disk): 43.27 MiB (45,374,364 bytes)
mAP@IoU=0.50 : 0.760824

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8m_1280_dynamic_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
FPS: 13.1065
device=gpu
Model size (on disk): 99.48 MiB (104,314,197 bytes)
mAP@IoU=0.50 : 0.757285
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

INT8 static quantization (GPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8s_320_static_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
2026-05-04 10:53:27.7155876 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime:MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 51.9161
device=gpu
Model size (on disk): 11.02 MiB (11,554,684 bytes)
mAP@IoU=0.50 : 0.564533
```

```

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8n_320_static_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
2026-05-04 10:53:10.8415642 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 57.1072
device=gpu
Model size (on disk): 3.31 MiB (3,474,227 bytes)
mAP@IoU=0.50 : 0.505161

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8m_320_static_quantized.onnx" --device gpu --input-size 320 --data-root ".\imagedata"
2026-05-04 10:53:45.9383275 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 44.1233
device=gpu
Model size (on disk): 25.22 MiB (26,448,568 bytes)
mAP@IoU=0.50 : 0.635499

```

640 input size

```

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8n_640_static_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
2026-05-04 10:54:09.7148065 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 39.3980
device=gpu
Model size (on disk): 3.43 MiB (3,600,269 bytes)
mAP@IoU=0.50 : 0.658794

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8s_640_static_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
2026-05-04 10:54:27.5805485 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 37.0933
device=gpu
Model size (on disk): 11.14 MiB (11,680,726 bytes)
mAP@IoU=0.50 : 0.684049

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8m_640_static_quantized.onnx" --device gpu --input-size 640 --data-root ".\imagedata"
2026-05-04 10:54:43.5697818 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 30.2650
device=gpu
Model size (on disk): 25.34 MiB (26,574,624 bytes)
mAP@IoU=0.50 : 0.708243

```

1280 input size

```

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8n_1280_static_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
2026-05-04 10:55:22.0742295 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 16.6463
device=gpu
Model size (on disk): 3.91 MiB (4,104,411 bytes)
mAP@IoU=0.50 : 0.710115

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8s_1280_static_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
2026-05-04 10:55:46.0572964 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 63 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 14.3319
device=gpu
Model size (on disk): 11.62 MiB (12,184,868 bytes)
mAP@IoU=0.50 : 0.733685

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8m_1280_static_quantized.onnx" --device gpu --input-size 1280 --data-root ".\imagedata"
2026-05-04 10:56:06.9808738 [W:onnxruntime:, transformer_memcpy.cc:111 onnxruntime::MemcpyTransformer::ApplyImpl] 83 Mempcy nodes are added to the graph main_graph for CUDAEExecutionProvider. It might have negative impact on performance (including unable to run CUDA graph). Set session_options.log_severity_level=1 to see the detail logs before this message.
FPS: 11.0798
device=gpu
Model size (on disk): 25.82 MiB (27,078,806 bytes)
mAP@IoU=0.50 : 0.747494

```

FP32 (CPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8n_320.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 67.3956
device=cpu
Model size (on disk): 12.14 MiB (12,725,005 bytes)
mAP@IoU=0.50 : 0.521438
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8s_320.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 47.6566
device=cpu
Model size (on disk): 42.67 MiB (44,743,799 bytes)
mAP@IoU=0.50 : 0.614228
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\320\yolov8m_320.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 29.9098
device=cpu
Model size (on disk): 98.88 MiB (103,683,476 bytes)
mAP@IoU=0.50 : 0.660906
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8n_640.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 27.9382
device=cpu
Model size (on disk): 12.26 MiB (12,851,087 bytes)
mAP@IoU=0.50 : 0.679877
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8s_640.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 19.3220
device=cpu
Model size (on disk): 42.79 MiB (44,869,881 bytes)
mAP@IoU=0.50 : 0.707885
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\640\yolov8m_640.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 10.9177
device=cpu
Model size (on disk): 99.00 MiB (103,809,586 bytes)
mAP@IoU=0.50 : 0.723569
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8n_1280.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 7.9756
device=cpu
Model size (on disk): 12.74 MiB (13,355,362 bytes)
mAP@IoU=0.50 : 0.725776
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8s_1280.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 5.2124
device=cpu
Model size (on disk): 43.27 MiB (45,374,156 bytes)
mAP@IoU=0.50 : 0.760904
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP32_onnx_models\1280\yolov8m_1280.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 2.8821
device=cpu
Model size (on disk): 99.48 MiB (104,313,941 bytes)
mAP@IoU=0.50 : 0.757325
```

Fp16 quantization (CPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8n_320_fp16.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 57.9423
device=cpu
Model size (on disk): 6.08 MiB (6,373,145 bytes)
mAP@IoU=0.50 : 0.520132
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8s_320_fp16.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 43.4874
device=cpu
Model size (on disk): 21.35 MiB (22,382,527 bytes)
mAP@IoU=0.50 : 0.614328
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\320\yolov8m_320_fp16.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 28.3625
device=cpu
Model size (on disk): 49.45 MiB (51,854,796 bytes)
mAP@IoU=0.50 : 0.663455
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8n_640_fp16.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 22.2073
device=cpu
Model size (on disk): 6.14 MiB (6,436,157 bytes)
mAP@IoU=0.50 : 0.679810
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8s_640_fp16.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 16.2747
device=cpu
Model size (on disk): 21.41 MiB (22,445,539 bytes)
mAP@IoU=0.50 : 0.707883
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\640\yolov8m_640_fp16.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 9.5101
device=cpu
Model size (on disk): 49.51 MiB (51,917,808 bytes)
mAP@IoU=0.50 : 0.723518
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8n_1280_fp16.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 6.3933
device=cpu
Model size (on disk): 6.38 MiB (6,688,163 bytes)
mAP@IoU=0.50 : 0.725676
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8s_1280_fp16.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 4.3144
device=cpu
Model size (on disk): 21.65 MiB (22,697,545 bytes)
mAP@IoU=0.50 : 0.761036
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\FP16_quantized_models\1280\yolov8m_1280_fp16.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 2.3095
device=cpu
Model size (on disk): 49.75 MiB (52,169,814 bytes)
mAP@IoU=0.50 : 0.758460
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

INT8 dynamic quantization (CPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8n_320_dynamic_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 64.7625
device=cpu
Model size (on disk): 12.14 MiB (12,725,187 bytes)
mAP@IoU=0.50 : 0.521438
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8s_320_dynamic_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 47.6578
device=cpu
Model size (on disk): 42.67 MiB (44,744,007 bytes)
mAP@IoU=0.50 : 0.614228
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\320\yolov8m_320_dynamic_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 30.7470
device=cpu
Model size (on disk): 98.88 MiB (103,683,732 bytes)
mAP@IoU=0.50 : 0.660906
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8n_640_dynamic_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 28.0307
device=cpu
Model size (on disk): 12.26 MiB (12,851,269 bytes)
mAP@IoU=0.50 : 0.679877
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8s_640_dynamic_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 19.5231
device=cpu
Model size (on disk): 42.79 MiB (44,870,089 bytes)
mAP@IoU=0.50 : 0.707885
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\640\yolov8m_640_dynamic_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 11.1965
device=cpu
Model size (on disk): 99.00 MiB (103,809,842 bytes)
mAP@IoU=0.50 : 0.723569
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8n_1280_dynamic_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 7.5783
device=cpu
Model size (on disk): 12.74 MiB (13,355,544 bytes)
mAP@IoU=0.50 : 0.725776
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8s_1280_dynamic_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 5.0016
device=cpu
Model size (on disk): 43.27 MiB (45,374,364 bytes)
mAP@IoU=0.50 : 0.760904
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_dynamic_quantized_models\1280\yolov8m_1280_dynamic_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 2.5723
device=cpu
Model size (on disk): 99.48 MiB (104,314,197 bytes)
mAP@IoU=0.50 : 0.757325
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

INT8 Static quantization (CPU2)

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8n_320_static_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 48.3574
device=cpu
Model size (on disk): 3.31 MiB (3,474,227 bytes)
mAP@IoU=0.50 : 0.497910
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8s_320_static_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 29.5353
device=cpu
Model size (on disk): 11.02 MiB (11,554,684 bytes)
mAP@IoU=0.50 : 0.564358
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\320\yolov8m_320_static_quantized.onnx" --device cpu --input-size 320 --data-root ".\imagedata"
FPS: 16.4254
device=cpu
Model size (on disk): 25.22 MiB (26,448,568 bytes)
mAP@IoU=0.50 : 0.628468
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8n_640_static_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 18.7071
device=cpu
Model size (on disk): 3.43 MiB (3,600,269 bytes)
mAP@IoU=0.50 : 0.651404
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8s_640_static_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 11.5520
device=cpu
Model size (on disk): 11.14 MiB (11,680,726 bytes)
mAP@IoU=0.50 : 0.601228
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\640\yolov8m_640_static_quantized.onnx" --device cpu --input-size 640 --data-root ".\imagedata"
FPS: 6.2983
device=cpu
Model size (on disk): 25.34 MiB (26,574,624 bytes)
mAP@IoU=0.50 : 0.715222
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det>
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8n_1280_static_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 6.0194
device=cpu
Model size (on disk): 3.91 MiB (4,104,411 bytes)
mAP@IoU=0.50 : 0.704724
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8s_1280_static_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 3.3785
device=cpu
Model size (on disk): 11.62 MiB (12,184,868 bytes)
mAP@IoU=0.50 : 0.734559
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\P_Det> python .\code_files\main_execution.py --model "models\yolo_onnx_models\INT8_static_quantized_models\1280\yolov8m_1280_static_quantized.onnx" --device cpu --input-size 1280 --data-root ".\imagedata"
FPS: 1.7779
device=cpu
Model size (on disk): 25.82 MiB (27,078,806 bytes)
mAP@IoU=0.50 : 0.757150
```

PyTorch

FP32 CPU1

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device cpu --input-size 640 --data-root imagedata
FPS: 12.2788
device=cpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.660008
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device cpu --input-size 640 --data-root imagedata
FPS: 9.8645
device=cpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.697099
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device cpu --input-size 640 --data-root imagedata
FPS: 4.3070
device=cpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.698181
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project>
```

320input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device cpu --input-size 320 --data-root imagedata
FPS: 35.1332
device=cpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.518057
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device cpu --input-size 320 --data-root imagedata
FPS: 26.2342
device=cpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.607383
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device cpu --input-size 320 --data-root imagedata
FPS: 12.7656
device=cpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.653156
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device cpu --input-size 1280 --data-root imagedata
FPS: 2.7130
device=cpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.718343
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device cpu --input-size 1280 --data-root imagedata
FPS: 1.6374
device=cpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.754759
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device cpu --input-size 1280 --data-root imagedata
FPS: 0.8124
device=cpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.748083
```

FP32 GPU1

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device gpu --input-size 640 --data-root imagedata
FPS: 49.5332
device=gpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.660611
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device gpu --input-size 640 --data-root imagedata
FPS: 44.7074
device=gpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.697108
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device gpu --input-size 640 --data-root imagedata
FPS: 24.0304
device=gpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.699283
```

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device gpu --input-size 320 --data-root imagedata
FPS: 60.7908
device=gpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.519373
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device gpu --input-size 320 --data-root imagedata
FPS: 58.4386
device=gpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.607374
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device gpu --input-size 320 --data-root imagedata
FPS: 37.4549
device=gpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.653158
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device gpu --input-size 1280 --data-root imagedata
FPS: 30.4704
device=gpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.718351
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device gpu --input-size 1280 --data-root imagedata
FPS: 19.1722
device=gpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.754749
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device gpu --input-size 1280 --data-root imagedata
FPS: 7.9640
device=gpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.748035
```

FP32 CPU2

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device cpu --input-size 640 --data-root imagedata
FPS: 31.8648
device=cpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.660008
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device cpu --input-size 640 --data-root imagedata
FPS: 24.0790
device=cpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.697099
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device cpu --input-size 640 --data-root imagedata
FPS: 13.2728
device=cpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.698181
```

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device cpu --input-size 320 --data-root imagedata
FPS: 55.8098
device=cpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.518057
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device cpu --input-size 320 --data-root imagedata
FPS: 44.2019
device=cpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.607383
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device cpu --input-size 320 --data-root imagedata
FPS: 27.0851
device=cpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.653156
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device cpu --input-size 1280 --data-root imagedata
FPS: 14.5623
device=cpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.718343
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device cpu --input-size 1280 --data-root imagedata
FPS: 8.7038
device=cpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.754759
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device cpu --input-size 1280 --data-root imagedata
FPS: 4.0792
device=cpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.748083
```

FP32 GPU2

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device gpu --input-size 640 --data-root imagedata
FPS: 63.4439
device=gpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.660001
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device gpu --input-size 640 --data-root imagedata
FPS: 62.4497
device=gpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.697097
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device gpu --input-size 640 --data-root imagedata
FPS: 57.7331
device=gpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.699277
```

320 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device gpu --input-size 320 --data-root imagedata
FPS: 61.4868
device=gpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.518066
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device gpu --input-size 320 --data-root imagedata
FPS: 61.9984
device=gpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.607371
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device gpu --input-size 320 --data-root imagedata
FPS: 55.0684
device=gpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.653148
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8n.pt --device gpu --input-size 1280 --data-root imagedata
FPS: 43.3629
device=gpu
Model size (on disk): 6.23 MiB (6,534,387 bytes)
mAP@IoU=0.50 : 0.718351
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8s.pt --device gpu --input-size 1280 --data-root imagedata
FPS: 41.3102
device=gpu
Model size (on disk): 21.53 MiB (22,573,363 bytes)
mAP@IoU=0.50 : 0.754744
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\yolo_models\yolov8m.pt --device gpu --input-size 1280 --data-root imagedata
FPS: 29.0190
device=gpu
Model size (on disk): 49.70 MiB (52,117,635 bytes)
mAP@IoU=0.50 : 0.748035
```

FP16 CPU1

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8n_640_fp16.pt --base-weights models\yolo_models\yolov8n.pt --device cpu --input-size 640 --data-root imagedata
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8n_640_fp16.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 22.9402
device=cpu
Model size (on disk): 6.25 MiB (6,549,510 bytes)
mAP@IoU=0.50 : 0.660008
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py --model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8s_640_fp16.pt --base-weights models\yolo_models\yolov8s.pt --device cpu --input-size 640 --data-root imagedata
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8s_640_fp16.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 10.0075
device=cpu
Model size (on disk): 21.54 MiB (22,588,550 bytes)
mAP@IoU=0.50 : 0.697099
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py -
-model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8m_640_fp16.pt --base-weights
models\yolo_models\yolov8m.pt --device cpu --input-size 640 --data-root imagedata
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\FP16_pytorch
_models\640\yolov8m_640_fp16.pt' appears to be improperly saved or formatted. For optimal results
, use model.save('filename.pt') to correctly save YOLO models.
FPS: 3.3327
device=cpu
Model size (on disk): 49.72 MiB (52,137,614 bytes)
mAP@IoU=0.50 : 0.698181
```

FP16 GPU1

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py -
• -model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8n_640_fp16.pt --base-weights
models\yolo_models\yolov8n.pt --device gpu --input-size 640 --data-root imagedata
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\FP16_pytorch
_models\640\yolov8n_640_fp16.pt' appears to be improperly saved or formatted. For optimal results
, use model.save('filename.pt') to correctly save YOLO models.
FPS: 52.8538
device=gpu
Model size (on disk): 6.25 MiB (6,549,510 bytes)
mAP@IoU=0.50 : 0.659246
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py -
• -model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8s_640_fp16.pt --base-weights
models\yolo_models\yolov8s.pt --device gpu --input-size 640 --data-root imagedata
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\FP16_pytorch
_models\640\yolov8s_640_fp16.pt' appears to be improperly saved or formatted. For optimal results
, use model.save('filename.pt') to correctly save YOLO models.
FPS: 47.6628
device=gpu
Model size (on disk): 21.54 MiB (22,588,550 bytes)
mAP@IoU=0.50 : 0.697073
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py -
• -model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8m_640_fp16.pt --base-weights
models\yolo_models\yolov8m.pt --device gpu --input-size 640 --data-root imagedata
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\FP16_pytorch
_models\640\yolov8m_640_fp16.pt' appears to be improperly saved or formatted. For optimal results
, use model.save('filename.pt') to correctly save YOLO models.
FPS: 32.7301
device=gpu
Model size (on disk): 49.72 MiB (52,137,614 bytes)
mAP@IoU=0.50 : 0.697034
```

FP16 CPU2

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8n_640_fp16.pt --base-weights models\yolo_models\yolov8n.pt --device cpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8n_640_fp16.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 37.3343
device=cpu
Model size (on disk): 6.25 MiB (6,549,510 bytes)
mAP@IoU=0.50 : 0.660008
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8s_640_fp16.pt --base-weights models\yolo_models\yolov8s.pt --device cpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8s_640_fp16.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 24.2299
device=cpu
Model size (on disk): 21.54 MiB (22,588,550 bytes)
mAP@IoU=0.50 : 0.697099
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8m_640_fp16.pt --base-weights models\yolo_models\yolov8m.pt --device cpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8m_640_fp16.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 12.4201
device=cpu
Model size (on disk): 49.72 MiB (52,137,614 bytes)
mAP@IoU=0.50 : 0.698181
```

FP16 GPU2

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8n_640_fp16.pt --base-weights models\yolo_models\yolov8n.pt --device gpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8n_640_fp16.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 55.0658
device=gpu
Model size (on disk): 6.25 MiB (6,549,510 bytes)
mAP@IoU=0.50 : 0.659246
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8s_640_fp16.pt --base-weights models\yolo_models\yolov8s.pt --device gpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8s_640_fp16.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 58.4418
device=gpu
Model size (on disk): 21.54 MiB (22,588,550 bytes)
mAP@IoU=0.50 : 0.697065
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmarking\main_execution.py --model models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8m_640_fp16.pt --base-weights models\yolo_models\yolov8m.pt --device gpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\FP16_pytorch_models\640\yolov8m_640_fp16.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 46.6650
device=gpu
Model size (on disk): 49.72 MiB (52,137,614 bytes)
mAP@IoU=0.50 : 0.698218
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project>
```

INT8 dynamic quantized CPU1

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py -  
-model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8n_640_pyto  
rch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8n.pt --device cpu --input-size 6  
40 --data-root imagedata  
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\INT8_pytorch  
_dynamic_quantized_models\640\yolov8n_640_pytorch_dynamic_quantized.pt' appears to be improperly  
saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO mod  
els.  
FPS: 21.6041  
device=cpu  
Model size (on disk): 12.34 MiB (12,944,567 bytes)  
mAP@IoU=0.50 : 0.660008  
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py -  
-model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8s_640_pyto  
rch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8s.pt --device cpu --input-size 6  
40 --data-root imagedata  
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\INT8_pytorch  
_dynamic_quantized_models\640\yolov8s_640_pytorch_dynamic_quantized.pt' appears to be improperly  
saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO mod  
els.  
FPS: 10.3536  
device=cpu  
Model size (on disk): 42.93 MiB (45,020,023 bytes)  
mAP@IoU=0.50 : 0.697099  
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py -  
-model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8m_640_pyto  
rch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8m.pt --device cpu --input-size 6  
40 --data-root imagedata  
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\INT8_pytorch  
_dynamic_quantized_models\640\yolov8m_640_pytorch_dynamic_quantized.pt' appears to be improperly  
saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO mod  
els.  
FPS: 4.3402  
device=cpu  
Model size (on disk): 99.25 MiB (104,070,615 bytes)  
mAP@IoU=0.50 : 0.698181  
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project>
```

INT8 dynamic quantized GPU1

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py -
-model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8n_640_pyto
rch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8n.pt --device gpu --input-size 6
40 --data-root imagedata
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\INT8_pytorch
_dynamic_quantized_models\640\yolov8n_640_pytorch_dynamic_quantized.pt' appears to be improperly
saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO mod
els.
FPS: 56.2006
device=gpu
Model size (on disk): 12.34 MiB (12,944,567 bytes)
mAP@IoU=0.50 : 0.660611
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py -
-model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8s_640_pyto
rch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8s.pt --device gpu --input-size 6
40 --data-root imagedata
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\INT8_pytorch
_dynamic_quantized_models\640\yolov8s_640_pytorch_dynamic_quantized.pt' appears to be improperly
saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO mod
els.
FPS: 47.2267
device=gpu
Model size (on disk): 42.93 MiB (45,020,023 bytes)
mAP@IoU=0.50 : 0.697108
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python pytorch_benchmarking\main_execution.py -
-model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8m_640_pyto
rch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8m.pt --device gpu --input-size 6
40 --data-root imagedata
WARNING The file 'C:\Users\rjrao\Desktop\605_Project\models\pytorch_quantized_models\INT8_pytorch
_dynamic_quantized_models\640\yolov8m_640_pytorch_dynamic_quantized.pt' appears to be improperly
saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO mod
els.
FPS: 24.2137
device=gpu
Model size (on disk): 99.25 MiB (104,070,615 bytes)
mAP@IoU=0.50 : 0.699283
```

INT8 dynamic quantized CPU2

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmark\main_execution.py --model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8n_640_pytorch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8n.pt --device cpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8n_640_pytorch_dynamic_quantized.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 36.5678
device=cpu
Model size (on disk): 12.34 MiB (12,944,567 bytes)
mAP@IoU=0.50 : 0.660008
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmark\main_execution.py --model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8s_640_pytorch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8s.pt --device cpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8s_640_pytorch_dynamic_quantized.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 24.7752
device=cpu
Model size (on disk): 42.93 MiB (45,020,023 bytes)
mAP@IoU=0.50 : 0.697099
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmark\main_execution.py --model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8m_640_pytorch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8m.pt --device cpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8m_640_pytorch_dynamic_quantized.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 11.8060
device=cpu
Model size (on disk): 99.25 MiB (104,070,615 bytes)
mAP@IoU=0.50 : 0.698181
```

INT8 dynamic quantized GPU2

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmark\main_execution.py --model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8n_640_pytorch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8n.pt --device gpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8n_640_pytorch_dynamic_quantized.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 64.6652
device=gpu
Model size (on disk): 12.34 MiB (12,944,567 bytes)
mAP@IoU=0.50 : 0.660001
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmark\main_execution.py --model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8s_640_pytorch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8s.pt --device gpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8s_640_pytorch_dynamic_quantized.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 61.9458
device=gpu
Model size (on disk): 42.93 MiB (45,020,023 bytes)
mAP@IoU=0.50 : 0.697097
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\pytorch_benchmark\main_execution.py --model models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8m_640_pytorch_dynamic_quantized.pt --base-weights models\yolo_models\yolov8m.pt --device gpu --input-size 640 --data-root imagedata
WARNING The file 'F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project\models\pytorch_quantized_models\INT8_pytorch_dynamic_quantized_models\640\yolov8m_640_pytorch_dynamic_quantized.pt' appears to be improperly saved or formatted. For optimal results, use model.save('filename.pt') to correctly save YOLO models.
FPS: 55.4476
device=gpu
Model size (on disk): 99.25 MiB (104,070,615 bytes)
mAP@IoU=0.50 : 0.699277
```

TensorRT GPU1

FP32

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tensorrt_benchmarking\main_execution.py --engine
models\tensorrt_quantized_models\FP32_trt_models\640\yolov8n_640_trt_fp32.engine --input-size 640 --data-r
oot imagedata
FPS: 52.1043
device=gpu
Model size (on disk): 17.96 MiB (18,836,980 bytes)
mAP@IoU=0.50 : 0.679877
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tensorrt_benchmarking\main_execution.py --engine
models\tensorrt_quantized_models\FP32_trt_models\640\yolov8s_640_trt_fp32.engine --input-size 640 --data-r
oot imagedata
FPS: 49.6425
device=gpu
Model size (on disk): 61.19 MiB (64,164,436 bytes)
mAP@IoU=0.50 : 0.707869
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tensorrt_benchmarking\main_execution.py --engine
models\tensorrt_quantized_models\FP32_trt_models\640\yolov8m_640_trt_fp32.engine --input-size 640 --data-r
oot imagedata
FPS: 35.1137
device=gpu
Model size (on disk): 148.04 MiB (155,229,364 bytes)
mAP@IoU=0.50 : 0.723540
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> 
```

FP16

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tensorrt_benchmarking\main_execution.py --engine
models\tensorrt_quantized_models\FP16_trt_models\640\yolov8n_640_trt_fp16.engine --input-size 640 --data-r
oot imagedata
FPS: 57.9883
device=gpu
Model size (on disk): 7.95 MiB (8,340,564 bytes)
mAP@IoU=0.50 : 0.679762
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tensorrt_benchmarking\main_execution.py --engine
models\tensorrt_quantized_models\FP16_trt_models\640\yolov8s_640_trt_fp16.engine --input-size 640 --data-r
oot imagedata
FPS: 54.0923
device=gpu
Model size (on disk): 23.34 MiB (24,468,588 bytes)
mAP@IoU=0.50 : 0.707808
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tensorrt_benchmarking\main_execution.py --engine
models\tensorrt_quantized_models\FP16_trt_models\640\yolov8m_640_trt_fp16.engine --input-size 640 --data-r
oot imagedata
FPS: 49.8339
device=gpu
Model size (on disk): 52.20 MiB (54,736,700 bytes)
mAP@IoU=0.50 : 0.722321
```

INT8

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tensorrt_benchmarking\main_execution.py --engine
models\tensorrt_quantized_models\INT8_trt_models\640\yolov8n_640_trt_int8.engine --input-size 640 --data-root
oot imagedata
FPS: 57.9249
device=gpu
Model size (on disk): 4.55 MiB (4,767,716 bytes)
mAP@IoU=0.50 : 0.565726
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tensorrt_benchmarking\main_execution.py --engine
models\tensorrt_quantized_models\INT8_trt_models\640\yolov8s_640_trt_int8.engine --input-size 640 --data-root
oot imagedata
FPS: 54.4020
device=gpu
Model size (on disk): 12.46 MiB (13,069,628 bytes)
mAP@IoU=0.50 : 0.644273
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tensorrt_benchmarking\main_execution.py --engine
models\tensorrt_quantized_models\INT8_trt_models\640\yolov8m_640_trt_int8.engine --input-size 640 --data-root
oot imagedata
FPS: 56.5572
device=gpu
Model size (on disk): 27.17 MiB (28,487,268 bytes)
mAP@IoU=0.50 : 0.669644
```

TensorRT GPU2

FP32

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tensorrt_benchmarking\main_execution
.py --engine models\tensorrt_quantized_models\FP32_trt_models\640\yolov8n_640_trt_fp32.engine --input-size 640 --data-root imagedata
FPS: 57.4010
device=gpu
Model size (on disk): 15.68 MiB (16,439,036 bytes)
mAP@IoU=0.50 : 0.679890
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tensorrt_benchmarking\main_execution
.py --engine models\tensorrt_quantized_models\FP32_trt_models\640\yolov8s_640_trt_fp32.engine --input-size 640 --data-root imagedata
FPS: 54.0234
device=gpu
Model size (on disk): 56.08 MiB (58,808,668 bytes)
mAP@IoU=0.50 : 0.707870
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tensorrt_benchmarking\main_execution
.py --engine models\tensorrt_quantized_models\FP32_trt_models\640\yolov8m_640_trt_fp32.engine --input-size 640 --data-root imagedata
FPS: 47.3200
device=gpu
Model size (on disk): 117.09 MiB (122,775,444 bytes)
mAP@IoU=0.50 : 0.723541
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project>
```

FP16

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tensorrt_benchmarking\main_execution
.py --engine models\tensorrt_quantized_models\FP16_trt_models\640\yolov8n_640_trt_fp16.engine --input-size 640 --data-root imagedata
FPS: 58.4916
device=gpu
Model size (on disk): 8.60 MiB (9,020,572 bytes)
mAP@IoU=0.50 : 0.679939
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tensorrt_benchmarking\main_execution
.py --engine models\tensorrt_quantized_models\FP16_trt_models\640\yolov8s_640_trt_fp16.engine --input-size 640 --data-root imagedata
FPS: 57.5626
device=gpu
Model size (on disk): 24.35 MiB (25,528,132 bytes)
mAP@IoU=0.50 : 0.707869
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tensorrt_benchmarking\main_execution
.py --engine models\tensorrt_quantized_models\FP16_trt_models\640\yolov8m_640_trt_fp16.engine --input-size 640 --data-root imagedata
FPS: 52.5428
device=gpu
Model size (on disk): 52.53 MiB (55,076,964 bytes)
mAP@IoU=0.50 : 0.723565
```

INT8

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tensorrt_benchmarking\main_execution
.py --engine models\tensorrt_quantized_models\INT8_trt_models\640\yolov8n_640_trt_int8.engine --input-size 640 --data-root imagedata
FPS: 60.2051
device=gpu
Model size (on disk): 4.64 MiB (4,868,620 bytes)
mAP@IoU=0.50 : 0.570273
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tensorrt_benchmarking\main_execution
.py --engine models\tensorrt_quantized_models\INT8_trt_models\640\yolov8s_640_trt_int8.engine --input-size 640 --data-root imagedata
FPS: 59.3436
device=gpu
Model size (on disk): 12.56 MiB (13,170,684 bytes)
mAP@IoU=0.50 : 0.645651
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tensorrt_benchmarking\main_execution
.py --engine models\tensorrt_quantized_models\INT8_trt_models\640\yolov8m_640_trt_int8.engine --input-size 640 --data-root imagedata
FPS: 57.9280
device=gpu
Model size (on disk): 27.67 MiB (29,017,004 bytes)
mAP@IoU=0.50 : 0.672744
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project>
```

TFLite

FP32 CPU1

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\640\yolov8n_640_fp32.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:38:37.052047: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:38:37.853814: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 10.3313
device=cpu
Model size (on disk): 12.33 MiB (12,924,383 bytes)
mAP@IoU=0.50 : 0.679877
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\640\yolov8s_640_fp32.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:38:57.605876: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:38:58.484186: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 3.7642
device=cpu
Model size (on disk): 42.86 MiB (44,942,943 bytes)
mAP@IoU=0.50 : 0.707885
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\640\yolov8m_640_fp32.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:39:39.974918: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:39:40.839612: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 1.0272
device=cpu
Model size (on disk): 99.06 MiB (103,875,304 bytes)
mAP@IoU=0.50 : 0.723569
```

320 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\320\yolov8n_320_fp32.tflite --device cpu --input-size 320 --data-root imagedata
2026-05-11 00:53:15.453757: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-11 00:53:16.589344: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 36.7998
device=cpu
Model size (on disk): 12.18 MiB (12,773,183 bytes)
mAP@IoU=0.50 : 0.521438
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\320\yolov8s_320_fp32.tflite --device cpu --input-size 320 --data-root imagedata
2026-05-11 00:53:32.604418: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-11 00:53:33.533575: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 7.8105
device=cpu
Model size (on disk): 42.72 MiB (44,791,743 bytes)
mAP@IoU=0.50 : 0.614228
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\320\yolov8m_320_fp32.tflite --device cpu --input-size 320 --data-root imagedata
2026-05-11 00:54:23.660984: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-11 00:54:24.564418: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 2.6491
device=cpu
Model size (on disk): 98.92 MiB (103,724,103 bytes)
mAP@IoU=0.50 : 0.660906
```

1280 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\1280\yolov8n_1280_fp32.tflite --device cpu --input-size 1280 --data-root imagedata
2026-05-11 00:55:33.833566: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off , set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-11 00:55:34.781609: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off , set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 1.3860
device=cpu
Model size (on disk): 12.90 MiB (13,529,184 bytes)
mAP@IoU=0.50 : 0.725776
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\1280\yolov8s_1280_fp32.tflite --device cpu --input-size 1280 --data-root imagedata
2026-05-11 00:57:16.585394: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off , set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-11 00:57:17.447943: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off , set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 0.2758
device=cpu
Model size (on disk): 43.44 MiB (45,547,745 bytes)
mAP@IoU=0.50 : 0.760904
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\1280\yolov8m_1280_fp32.tflite --device cpu --input-size 1280 --data-root imagedata
2026-05-11 01:04:09.062778: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off , set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-11 01:04:09.980492: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off , set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 0.1624
device=cpu
Model size (on disk): 99.64 MiB (104,480,105 bytes)
mAP@IoU=0.50 : 0.757325
```

FP16 CPU1

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking/main_execution.py --model models\tflite_quantized_models\FP16_tflite_models\640\yolov8n_640_fp16.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:42:59.434352: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:43:00.319882: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 10.4949
device=cpu
Model size (on disk): 6.23 MiB (6,535,068 bytes)
mAP@IoU=0.50 : 0.679918
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking/main_execution.py --model models\tflite_quantized_models\FP16_tflite_models\640\yolov8s_640_fp16.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:43:20.630628: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:43:21.448451: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 3.7887
device=cpu
Model size (on disk): 21.50 MiB (22,544,349 bytes)
mAP@IoU=0.50 : 0.706397
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking/main_execution.py --model models\tflite_quantized_models\FP16_tflite_models\640\yolov8m_640_fp16.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:44:01.523665: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:44:02.362782: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 1.3239
device=cpu
Model size (on disk): 49.61 MiB (52,022,196 bytes)
mAP@IoU=0.50 : 0.723548
```

Dynamic Range CPU1

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking/main_execution.py --model models\tflite_quantized_models\dynamic_range_tflite_models\640\yolov8n_640_dynamic_range.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:47:57.430827: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:47:58.276673: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 19.1922
device=cpu
Model size (on disk): 3.30 MiB (3,455,976 bytes)
mAP@IoU=0.50 : 0.672558
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking/main_execution.py --model models\tflite_quantized_models\dynamic_range_tflite_models\640\yolov8s_640_dynamic_range.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:48:12.475403: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:48:13.332821: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 2.7843
device=cpu
Model size (on disk): 11.00 MiB (11,532,520 bytes)
mAP@IoU=0.50 : 0.706848
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking/main_execution.py --model models\tflite_quantized_models\dynamic_range_tflite_models\640\yolov8m_640_dynamic_range.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:49:11.788100: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:49:12.646877: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 3.7988
device=cpu
Model size (on disk): 25.15 MiB (26,372,888 bytes)
mAP@IoU=0.50 : 0.728069
```

Full integer INT8 CPU1

640 input size

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking/main_execution.py --model models\tflite_quantized_models\full_integer_INT8_tflite_models\640\yolov8n_640_int8_full_integer.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:57:07.118112: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:57:07.982598: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 4.5123
device=cpu
Model size (on disk): 3.39 MiB (3,557,160 bytes)
mAP@IoU=0.50 : 0.672558
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking/main_execution.py --model models\tflite_quantized_models\full_integer_INT8_tflite_models\640\yolov8s_640_int8_full_integer.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:57:41.483126: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:57:42.380056: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 6.9061
device=cpu
Model size (on disk): 11.09 MiB (11,633,704 bytes)
mAP@IoU=0.50 : 0.706848
```

```
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project> python tflite_benchmarking/main_execution.py --model models\tflite_quantized_models\full_integer_INT8_tflite_models\640\yolov8m_640_int8_full_integer.tflite --device cpu --input-size 640 --data-root imagedata
2026-05-10 22:58:11.858557: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
2026-05-10 22:58:12.740582: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\rjrao\anaconda3\envs\yolo_env\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in TF 2.20. Please use the LitERT interpreter from the ai_edge_litert package. See the [migration guide](https://ai.google.dev/edge/litert/migration) for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 3.2607
device=cpu
Model size (on disk): 25.25 MiB (26,473,912 bytes)
mAP@IoU=0.50 : 0.728069
(yolo_env) PS C:\Users\rjrao\Desktop\605_Project>
```

TFLite

FP32 CPU2

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\640\yolov8n_640_fp32.tflite --device cpu --input-size 640 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519193.789008    5720 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519196.356968    5720 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning:   Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 10.5576
device=cpu
Model size (on disk): 12.33 MiB (12,924,383 bytes)
mAP@IoU=0.50 : 0.679877
```

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\640\yolov8s_640_fp32.tflite --device cpu --input-size 640 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519223.487846    27068 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519225.795866    27068 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning:   Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 4.2676
device=cpu
Model size (on disk): 42.86 MiB (44,942,943 bytes)
mAP@IoU=0.50 : 0.707885
```

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\640\yolov8m_640_fp32.tflite --device cpu --input-size 640 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519263.932255    23720 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519266.296064    23720 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning:   Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 1.6020
device=cpu
Model size (on disk): 99.06 MiB (103,875,304 bytes)
mAP@IoU=0.50 : 0.723569
```

320 input size

```

(yolo_quant) PS F:\Final_Lattice_Data_Structure\Active\MixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tfLite_benchmarking\main_execution.py --model models\tfLite_quantized\models\FP32_tfLite_models\320\yolov8n_320_fp32.tflite --device cpu --input-size 320 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519362.019539 28320 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519364.341525 28320 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

```

```
warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 33.5914
device-cpu
Model size (on disk): 12.18 MiB (12,773,183 bytes)
mAP@IoU=0.50 : 0.521438
```

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tfLite_benchmarking\main_execution.py --model models\tfLite_quantized_models\FP32_tflite_models\320\yolov8s_320_fp32.tflite --device cpu --input-size 320 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:17.78519417.053583 8772 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:17.78519419.414155 8772 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
C:\Users\suratirao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
See the [migration guide](https://ai.google.dev/edge/litert/migration)
for details.
```

```
warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 15.0384
device=cpu
Model size (on disk): 42.72 MiB (44,791,743 bytes)
mAP@IoU=0.50 : 0.614228
```

```
(yolo_quant) PS F:\Final Lattice Data Structure\Active\MixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tf_lite_benchmarking\main_execution.py --model models\tf_lite_quantized\models\FP32_tf_lite_models\320\yolov8m_320_fp32_tf_lite --device cpu --input-size 320 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519444.426847    26100 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519446.872211    26100 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
See the [migration guide](https://ai.google.dev/edge/litert/migration)
for details.
```

```
warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 5.9494
device-cpu
Model size (on disk): 98.92 MiB (103,724,103 bytes)
mAP@IoU=0.50 : 0.660906
```

1280 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tfllite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\1280\yolov8n_1280_fp32.tflite --device cpu --input-size 1280 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519483.432725 31360 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519485.758590 31360 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 2.6770
device=cpu
Model size (on disk): 12.90 MiB (13,529,184 bytes)
mAP@IoU=0.50 : 0.725776

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tfllite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\1280\yolov8s_1280_fp32.tflite --device cpu --input-size 1280 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519536.743985 14056 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519539.060559 14056 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 1.0431
device=cpu
Model size (on disk): 43.44 MiB (45,547,745 bytes)
mAP@IoU=0.50 : 0.760904

(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tfllite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP32_tflite_models\1280\yolov8m_1280_fp32.tflite --device cpu --input-size 1280 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519654.860011 29524 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519657.102000 29524 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

  warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 0.4073
device=cpu
Model size (on disk): 99.64 MiB (104,480,105 bytes)
mAP@IoU=0.50 : 0.757325
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project>
```

FP16 CPU2

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tfllite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP16_tflite_models\640\yolov8n_640_fp16.tflite --device cpu --input-size 640 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519969.488070    17876 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778519971.873769    17876 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 10.6131
device=cpu
Model size (on disk): 6.23 MiB (6,535,068 bytes)
mAP@IoU=0.50 : 0.679913
```

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tfllite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP16_tflite_models\640\yolov8s_640_fp16.tflite --device cpu --input-size 640 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520011.603363    32324 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520013.907840    32324 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 4.2973
device=cpu
Model size (on disk): 21.50 MiB (22,544,349 bytes)
mAP@IoU=0.50 : 0.706397
```

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tfllite_benchmarking\main_execution.py --model models\tflite_quantized_models\FP16_tflite_models\640\yolov8m_640_fp16.tflite --device cpu --input-size 640 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520048.135645    21396 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520050.492190    21396 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 1.6664
device=cpu
Model size (on disk): 49.61 MiB (52,022,196 bytes)
mAP@IoU=0.50 : 0.723548
```

Dynamic Range CPU2

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tf_lite_benchmarking\main_execution.py --model models\tf_lite_quantized_models\dynamic_range_tf_lite_models\640\yolov8n_640_dynamic_range.tf_lite --device cpu --input-size 640 --data-root images\agedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520145.422264    26268 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520147.686362    26268 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 20.1043
device=cpu
Model size (on disk): 3.30 MiB (3,455,976 bytes)
mAP@IoU=0.50 : 0.676014
```

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tf_lite_benchmarking\main_execution.py --model models\tf_lite_quantized_models\dynamic_range_tf_lite_models\640\yolov8s_640_dynamic_range.tf_lite --device cpu --input-size 640 --data-root images\agedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520160.442213    14020 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520162.760566    14020 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 10.8215
device=cpu
Model size (on disk): 11.00 MiB (11,532,520 bytes)
mAP@IoU=0.50 : 0.710357
```

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tf_lite_benchmarking\main_execution.py --model models\tf_lite_quantized_models\dynamic_range_tf_lite_models\640\yolov8m_640_dynamic_range.tf_lite --device cpu --input-size 640 --data-root images\agedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520186.695107    9292 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520189.000903    9292 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 5.4915
device=cpu
Model size (on disk): 25.15 MiB (26,372,888 bytes)
mAP@IoU=0.50 : 0.728225
```

Full integer INT8 CPU2

640 input size

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\full_integer_INT8_tflite_models\640\yolov8n_640_int8_full_integer.tflite --device cpu --input-size 640 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520253.524938 30704 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520255.833543 30704 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 19.8451
device=cpu
Model size (on disk): 3.39 MiB (3,557,160 bytes)
mAP@IoU=0.50 : 0.676014
```

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\full_integer_INT8_tflite_models\640\yolov8s_640_int8_full_integer.tflite --device cpu --input-size 640 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520271.918554 10136 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520274.266160 10136 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 10.8917
device=cpu
Model size (on disk): 11.09 MiB (11,633,704 bytes)
mAP@IoU=0.50 : 0.710357
```

```
(yolo_quant) PS F:\Final_Lattice_Data_Structure\ActiveMixedMaterialDataset\Suraj_Code_Files\605_Project> python .\tflite_benchmarking\main_execution.py --model models\tflite_quantized_models\full_integer_INT8_tflite_models\640\yolov8m_640_int8_full_integer.tflite --device cpu --input-size 640 --data-root imagedata
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520298.444792 22168 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1778520300.829529 22168 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
C:\Users\surajrao\AppData\Local\miniconda3\envs\yolo_quant\lib\site-packages\tensorflow\lite\python\interpreter.py:457: UserWarning: Warning: tf.lite.Interpreter is deprecated and is scheduled for deletion in
  TF 2.20. Please use the LiteRT interpreter from the ai_edge_litert package.
  See the [migration guide](https://ai.google.dev/edge/litert/migration)
  for details.

warnings.warn(_INTERPRETER_DELETION_WARNING)
INFO: Created TensorFlow Lite XNNPACK delegate for CPU.
FPS: 5.6383
device=cpu
Model size (on disk): 25.25 MiB (26,473,912 bytes)
mAP@IoU=0.50 : 0.728225
```